

High-Level Design (HLD)

1. System Overview

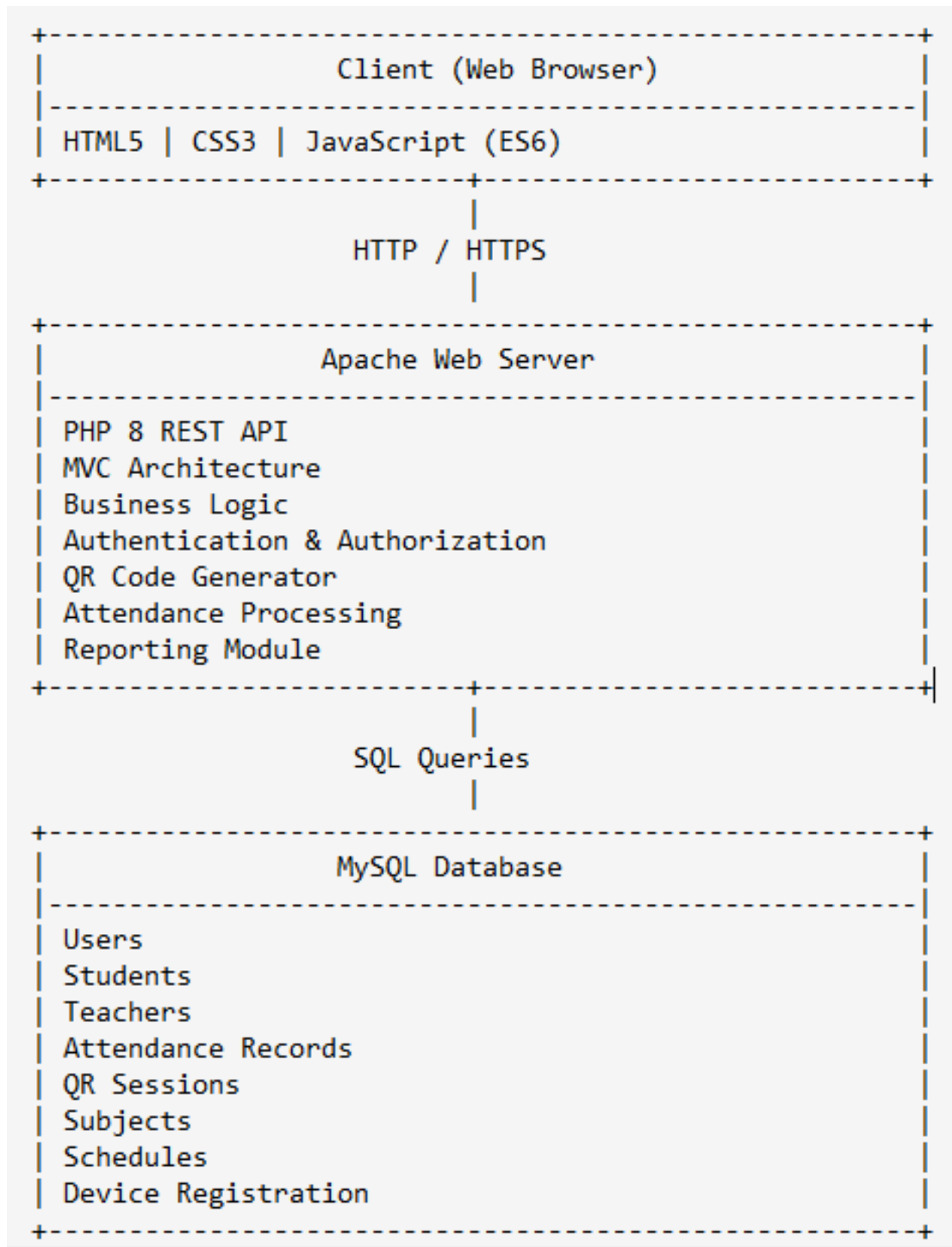
The **QR Code Attendance System** is a web-based application designed to provide a fast, secure, and efficient method for recording student attendance. The system replaces manual attendance with QR code scanning while preventing common issues such as proxy attendance and unauthorized account sharing.

The system consists of three user roles:

- **Administrator** – Manages users, schedules, classrooms, and system settings.
- **Teacher** – Creates attendance sessions, generates QR codes, monitors live attendance, and views attendance reports.
- **Student** – Scans QR codes to record attendance and views personal attendance records and schedules.

The application follows a **three-tier architecture** consisting of the presentation layer, application layer, and data layer.

2. System Architecture



3. Technology Stack

Layer	Technology
Frontend	HTML5, CSS3, JavaScript (ES6)
Backend	PHP 8
Architecture	REST API + MVC
Web Server	Apache
Database	MySQL Server

4. Major System Components

4.1 Presentation Layer (Frontend)

The presentation layer provides the user interface for all system users.

Responsibilities include:

- Login and authentication pages
- Teacher dashboard
- Student dashboard
- Administrator dashboard
- QR code display
- QR code scanner interface
- Attendance history
- Reports
- Schedule viewing
- Notifications

The frontend communicates with the backend through REST APIs.

4.2 Application Layer (Backend)

The backend processes all business logic.

Main responsibilities include:

- User authentication
- Device verification
- QR code generation
- QR validation
- Attendance recording
- Attendance percentage calculation
- Session management
- Schedule management
- Notification processing
- Report generation

The backend is implemented using **PHP 8** with the **MVC architecture**.

4.3 Database Layer

The database stores all persistent system information.

Major data includes:

- User accounts
 - Student information
 - Teacher information
 - Device registration
 - Attendance records
 - QR code sessions
 - Subjects
 - Class schedules
 - Semester records
-

5. Main Functional Modules

5.1 Authentication Module

Responsible for:

- User login
- User logout
- Role-based authentication
- Account verification
- Device restriction
- Session management

Features:

- Student accounts require administrator approval.
 - Student login is restricted to one registered device.
 - A unique UUID is generated during the first login and stored locally for device identification.
 - Device changes require administrator approval.
-

5.2 QR Attendance Module

Responsible for attendance collection.

Functions:

- Create attendance session
- Generate QR code
- Set QR expiration time
- Disable previous QR code when a new one is generated
- Validate QR scans
- Record attendance
- Manual attendance entry (optional)

The QR code remains valid only for the configured duration.

5.3 Attendance Management Module

Responsible for attendance processing.

Functions include:

- Record attendance
- Prevent duplicate attendance
- View attendance history
- Monthly attendance calculation
- Attendance percentage calculation
- Semester attendance summary

Teachers can immediately monitor:

- Total students
- Present students
- Absent students

through the live attendance dashboard.

5.4 Schedule Management Module

Teachers and administrators can manage class schedules.

Functions:

- Create schedules
- Save reusable attendance events
- Assign classrooms
- Assign subjects
- Set attendance month
- Manage semester periods

Students can view their schedules from their accounts.

5.5 User Management Module

Managed by the administrator.

Functions:

- Verify student accounts
 - Verify teacher accounts
 - Enable or disable accounts
 - Reset device registration
 - Manage student information
 - Manage teacher information
-

5.6 Notification Module

Provides important system notifications.

Examples include:

- Low attendance warnings
 - Important event announcements
 - Attendance reminders
 - QR session expiration notifications
-

5.7 Reporting Module

Provides attendance statistics.

Teachers can view:

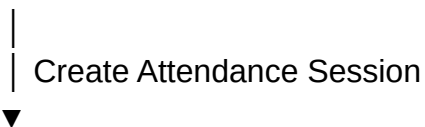
- Monthly attendance percentage
- Student attendance reports
- Subject attendance reports
- Semester summaries

Students can view:

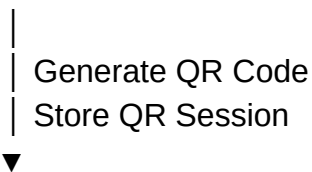
- Personal attendance history
 - Monthly attendance percentage
-

6. Component Interaction

Teacher

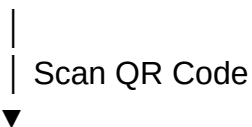


Backend API

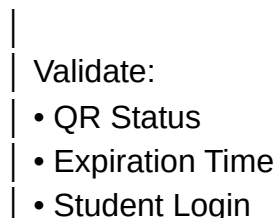


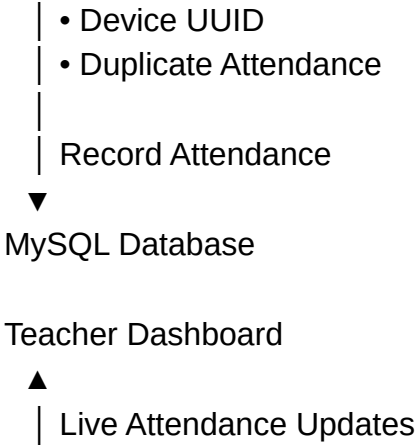
MySQL Database

Student



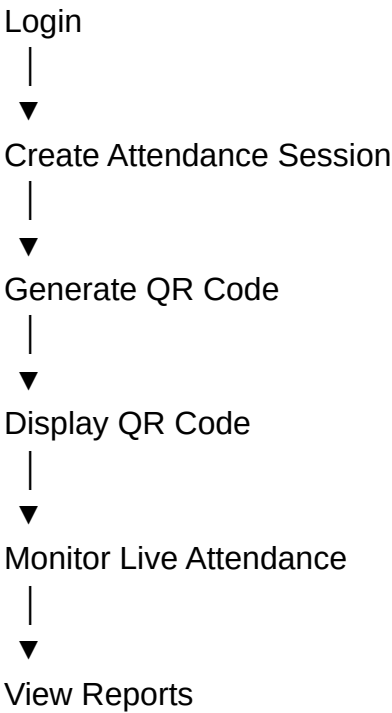
Backend API



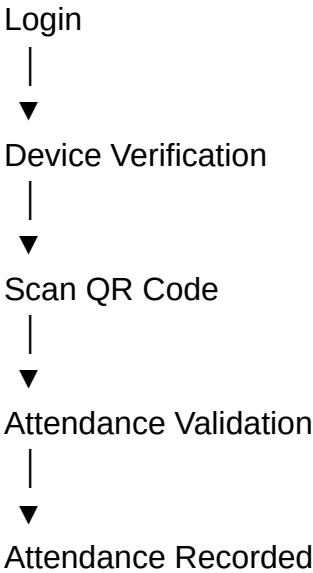


7. High-Level Data Flow

Teacher



Student



|
▼
View Attendance History

Administrator

Login
|
▼
Verify Accounts
|
▼
Manage Users
|
▼
Manage Schedules
|
▼
Reset Device Registration
|
▼
Generate Reports

8. System Security Design

The system incorporates several security mechanisms:

- Role-based access control (Administrator, Teacher, Student)
 - Password authentication
 - REST API authorization
 - One registered device per student account using a locally stored UUID
 - Administrator approval for account registration
 - QR code expiration after a configurable duration
 - Automatic invalidation of previous QR codes when a new one is generated
 - Prevention of duplicate attendance submissions
 - Secure session management
-

9. Future Enhancements

The system is designed to support future extensions, including:

- GPS-based classroom location verification
- Face recognition during attendance
- Push notifications for attendance reminders
- Check-in and check-out attendance tracking
- Permanent QR mode for special events
- Mobile application integration

- Analytics dashboard with attendance trends
- Export attendance reports to PDF and Excel

Low-Level Design (LLD)

1. Module Design

1.1 Authentication Module

Purpose

Authenticate users and manage user sessions.

Functions

- Login
- Logout
- Password verification
- Session creation
- Role identification
- Device verification

Inputs

- Username
- Password

Outputs

- Login success/failure
- User session
- JWT/Session ID (if applicable)

1.2 Student Module

Purpose

Manage student-related operations.

Functions

- View profile
- Scan QR code
- View attendance history
- View monthly attendance percentage
- View class schedule

Database Tables

- students
- attendance
- schedules

1.3 Teacher Module

Purpose

Manage attendance sessions.

Functions

- Create attendance event
- Generate QR code
- Set QR expiration time
- View live attendance
- View reports
- Manual attendance entry

Database Tables

- attendance_sessions
- attendance
- subjects
- schedules

1.4 Administrator Module

Purpose

Manage system users and settings.

Functions

- Verify student accounts
- Verify teacher accounts
- Enable/Disable users
- Reset device registration
- Manage schedules
- Manage classrooms

1.5 QR Code Module

Purpose

Generate and validate QR codes.

Functions

- Generate QR token
- Save QR session
- Validate QR token
- Check expiration time
- Disable previous QR

1.6 Attendance Module

Purpose

Record attendance.

Functions

- Validate student
- Validate QR
- Check duplicate attendance
- Save attendance
- Calculate attendance percentage

2. REST API Design

Authentication

Method	Endpoint	Description
POST	/api/login	User login
POST	/api/logout	Logout

Student APIs

Method	Endpoint	Description
GET	/api/student/profile	View profile
GET	/api/student/attendance	Attendance history
GET	/api/student/schedule	View schedule
POST	/api/student/scan	Submit attendance

Teacher APIs

Method	Endpoint	Description
POST	/api/attendance/create	Create attendance session
POST	/api/attendance/generateQR	Generate QR code
GET	/api/attendance/live	Live attendance
POST	/api/attendance/manual	Manual attendance
GET	/api/reports/monthly	Monthly report

Administrator APIs

Method	Endpoint	Description
POST	/api/admin/verify	Verify account
PUT	/api/admin/device/reset	Reset registered device
POST	/api/admin/schedule	Create schedule

3. Database Access

Module	Main Tables
Authentication	users
Student	students, attendance
Teacher	attendance_sessions, attendance
QR Module	qr_codes
Reports	attendance
Schedule	schedules

4. QR Attendance Process

Teacher clicks "Generate QR"



Create Attendance Session



Generate Unique QR Token



Store Token in Database



Display QR Code



Student Scans QR



Validate Token



Check Expiration



Check Student Login



Check Registered Device



Check Duplicate Attendance



Save Attendance



Update Teacher Dashboard

5. Login Process

User enters username/password



Validate Credentials



Check Account Status



Check Registered Device (Student)

|



Create Session

|



Redirect Dashboard

6. Attendance Percentage Calculation

Monthly attendance percentage is calculated using:

Attendance Percentage =

(Number of Classes Attended ÷

Total Number of Classes) × 100

Example:

Classes Conducted = 20

Student Attended = 18

Attendance = $(18 / 20) \times 100 = 90\%$

7. Validation Rules

Module	Validation
Login	Username and password required
Student Registration	Student card photo required, administrator verification required
Device Registration	One device UUID per student account
QR Generation	Expiration time must be greater than 0
QR Scan	QR token must exist and not be expired
Attendance	One attendance record per student per session
Manual Attendance	Teacher authorization required

8. Error Handling

Error	System Response
Invalid login	"Invalid username or password."
Account not verified	"Your account is pending administrator approval."
Device mismatch	"This account is registered on another device."
QR expired	"The QR code has expired."
Duplicate attendance	"Attendance has already been recorded."
Invalid QR	"Invalid QR code."
Unauthorized access	HTTP 401 Unauthorized
Forbidden action	HTTP 403 Forbidden

